

[Supplemental] Common data storage systems

Data analysts are responsible for gathering and organizing data. Depending on the type of data that you're working with, there are many ways in which it can be stored. Two systems for storing data are relational databases and NoSQL databases. In this reading, you'll learn about the differences between relational databases and NoSQL databases.

Data storage systems

Relational databases and NoSQL differ in how they structure stored data, and how data analysts interact with them. In order to interact with data in a database, analysts write queries. A query is a statement written in a programming language that accesses or manipulates data in a database. The programming language that data analysts use depends on what they're trying to accomplish, and how the database is structured.

Relational databases

Relational databases use a set schema to organize data within one or more tables that can be connected to form relationships. A schema is a description of how a database is organized. In relational databases, data is organized into tables by rows and columns. Each row contains information for a single record, such as a single customer order for an ecommerce website. Columns contain information for unique attributes for each record, such as the customer's name, or the revenue or tracking number for the order.

The tables within a relational database are connected using primary and foreign keys. A primary key is an identifier that references a column in which each value is unique. For example, the table for ecommerce orders might include a column for unique order numbers, which can be used to identify specific rows in the table. A foreign key is a column within a table that is a primary key in another table. For example, the orders table might also include a customer ID column, which is the primary key in a customers table. Since the customer ID column is shared between both tables, it's a foreign key in the orders table.

When two different tables share at least one column, then they have a relationship. Data analysts often use relational databases, because they allow you to query information from multiple tables as long as the tables share at least one primary or foreign key. Data analysts use

SQL as the standard language for interacting with relational databases, which is why relational databases are also commonly referred to as SQL databases.

NoSQL databases

Data analysts can also use NoSQL databases, which do not structure data into tables and therefore don't depend on relationships to organize and access data. Instead, NoSQL databases use dynamic, unstructured data organization options that don't follow a set schema.

The unstructured organization of NoSQL databases make them useful for storing and modifying extremely large data sets, because no organization system has to be maintained. However, the lack of structure can also make it more difficult to query data. NoSQL is usually chosen to store large amounts of data that come from a variety of sources that don't align with a relational database schema. Because of their varied organization and level of structure, each NoSQL database may have its own programming language that allows data analysts to interact with the data. Social media companies often use NoSQL databases to manage large amounts of unstructured data, such as posts, likes, comments, and connections.

Key takeaways

Data analysts can use a variety of methods for storing and organizing data, but relational and NoSQL databases are two of the most common. Relational databases organize data by a set schema that enables analysts to access data using the SQL programming language. NoSQL databases allow data from different sources that are not organized by a set schema to be included within the same database. Each NoSQL database uses its own programming language that aligns with its data and structure. By familiarizing yourself with different programming languages and data management systems, you can become a more effective data analyst, and better adapt to different data types and analysis needs.